University of Barishal

Department of Computer Science and Engineering

Course Title: Discrete Mathematics

Course Code: CSE-1203

1st Year, 2nd Semester Final Examination

Admission Session: 2019-20

Time: 03 Hours **Marks:** 60

N.B.: Answer any **Five** questions out of the followings. All parts of each question must be answered consecutively. Right side of the question shows the maximum marks.

1.a) Suppose that Smartphone A has 256 MB RAM and 32 GB ROM, and the resolution of its camera is 8 MP; Smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP; and Smartphone C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5 MP. Determine the truth value of each of these propositions. 2

i) Smartphone B has the most RAM of these three smartphones.
ii) Smartphone C has more ROM or a higher resolution camera than Smartphone B.
iii) Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A.
iv) If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera.

b) Define propositional logic. Write down the difference between implication and biconditional with example. 4

c) Construct the truth table of the compound proposition: $(p \vee \neg q) \rightarrow (p \wedge q)$ 3

d) Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology. 3

2.a) Let $P(x)$ be the statement "$x=x^2$". If the domain consists of integers, what are these truth values? 3

i) $P(0)$ ii) $\exists x P(x)$ iii) $\forall x P(x)$

b) Let f1 and f2 be functions from **R** to **R** such that f1(x) = x2 and f2(x) = x – x2. What are the functions f1 + f2 and f1f2? 4

c) Mention some methods of proving theorems. Prove that if $m + n$ and $n + p$ are even integers, where m, n, and p are integers, then $m + p$ is even. What kind of proof did you use? 5

3.a) Use set builder notation to give a description of each of these sets: 3

i) {-3, -2, -1, 0, 1, 2, 3} ii) {m, n, o, p}

b) Show that $A \times A \neq B \times A$, when A and B are nonempty, unless A = B. 3

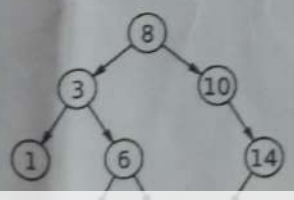
c) Define set with example. What is the power set of {a, b, c, 2} and Cartesian products of {1, 2, 3} and {x, y, z}. 4

d) Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself one-to-one. 2

4.a) Let $p(n) = a_0 + a_1 n + a_2 n^2 + \ldots + a_m n^m$; that is, suppose degree p(n) = m, prove that $p(n) = O(n^m)$. 4

b) Use mathematical induction to prove that $2^n < n!$ for every integer n with $n \geq 4$. (Note that this inequality is false for n = 1, 2, and 3.) 4

c) Analysis the complexity of the following binary search tree. 4



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5.a) Show that if n is an integer greater than 1, then n can be written as a prime or product primes. 4

b) Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? 3

c) Prove that, if k + 1 or more pigeons are distributed among k pigeonholes, then at least one pigeonhole contains two or more pigeons. 3

d) A bag contains 10 red marbles, 10 white marbles, and 10 blue marbles. What is the minimum no. of marbles you have to choose randomly from the bag to ensure that we get 4 marbles of same color? 2

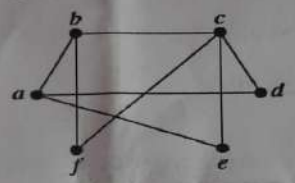
6.a) Suppose that A and B are events from a sample space S such that $P(A) \neq 0$ and $P(B) \neq 0$. Then proof the following: 5

$$p(B|A) = \frac{p(A \mid B)\, p(B)}{p(A \mid B)\, p(B) + P(A \mid \bar{B})\, p(\bar{B})}$$

b) What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5? 3

c) What do you mean by a complete graph? Draw a K_8 graph. What is the degree of a vertex of the K_8 graph you have drawn? 4

7.a) What is planner graph? Draw the planner graph of the given simple graph. 4



b) Explain Eulerian and Hamiltonian graphs with examples, also draw the graphs of the following: 4

i) Eulerian but not Hamiltonian ii) Hamiltonian but not Eulerian

c) Define graph coloring and chromatic number of a graph and find the chromatic number of

i) $K_{3,3}$ ii) cycle with even number of vertices

8.a) Suppose the pre-order and in-order traversals of a binary tree T yield the following sequence of nodes: 4

Pre-order: G, B, Q, A, C, K, F, P, D, E, R, H

In-order: Q, B, K, C, F, A, G, P, E, D, H, R

i) Draw the diagram of T ii) Find the depth d of T

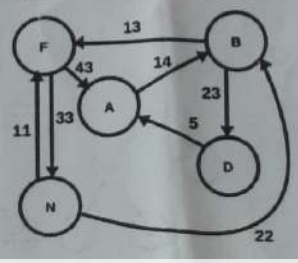
iii) Find the descendants of B iv) List the terminal nodes of T

b) Consider the following addresses which are in random order: 6

1, 2.2.1, 3.2, 2.2.1.1, 1.1.1, 0, 2.1, 3.2.1.1, 3, 3.1, 2.2, 2.1.1, 3.2.1, 1.1, 3.2.1.2, 2, 1.1.2

i) Place the addresses in lexicographic order ii) Draw the corresponding ordered rooted tree

c) Find the in-degree and out-degree of each vertex in the following graph with directed edges. 2



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